LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:16 Edge Detection Using Canny Method

Aim

To detect the edges in an image using the Canny Edge Detection method in Python and OpenCV.

Algorithm

1. Import the required libraries, cv2.
2. Read the input image.
3. Convert the image from BGR to grayscale.
4. Apply Gaussian Blur to reduce noise.
5. Apply the Canny Edge Detection method.
6. Display the original image and detected edges.
7. Press any key to close the output windows.

CODE:

EXP-16.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2

# Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Reduce noise using Gaussian Blur
blur = cv2.GaussianBlur(gray, (5, 5), 0)

# Apply Canny Edge Detection
edges = cv2.Canny(blur, 100, 200)

# Display original image
cv2.imshow("Original Image", image)

# Display edge detected image
cv2.imshow("Canny Edge Detection", edges)

# Wait for a key
cv2.waitKey(0)

# Close all windows
cv2.destroyAllWindows()
```

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OUTPUT :

Result:

Edge detection was successfully performed using the Canny Edge Detection method. The important boundaries and edges of objects in the image were detected and displayed.